LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 4: Swapping and the Bubble Pass

Module 2 | 45 minutes | Swapping and comparisons | Grades 6–8 · ages 12–14

Objective: Swap two list items safely with a temporary variable, and write one bubble-sort pass that moves the largest value to the end.

Success criteria

- I can swap two items with a temp variable.
- I can write one bubble pass.
- I can explain what a pass achieves.

Key vocabulary

- **swap**: Exchange two values.
- **temporary variable**: A variable that holds a value briefly.
- **pass**: One trip through the list comparing neighbours.
- **adjacent**: Next to each other.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Human bubble pass (5 min): five learners holding numbers; compare neighbours and swap places.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Swap, then one pass: learners follow the steps.
14 min	**Perform**	Code Lab: swap two items, Debug the swap
9 min	**Probe**	Code Lab: one bubble pass, activity
4 min	**Wrap-up**	Exit ticket: After one bubble pass, where is the largest value?

Unplugged warm-up

Human bubble pass (5 min): five learners holding numbers; compare neighbours and swap places.

Teaching the idea

Swapping requires a temporary store to avoid overwriting. A bubble pass compares adjacent pairs (n−1 comparisons), swapping out-of-order pairs, moving the maximum to the end. Procedures with a parameter (swap (i)) improve readability.

Analogy: Swapping two drinks needs a third cup.

Live demo: Show the broken swap (3, 3, 8, 1), then fix it with temp.

Questions to ask

Ask	Listen for
Why save to temp first?	Otherwise the value is overwritten.
Where is the largest value after one pass?	At the end.
How many comparisons for 10 items?	9.

Differentiation

Support
Provide swap (i); learners write the pass loop.

Stretch
Count the swaps as well as the comparisons.

Common misconceptions

Misconception	How to fix it
Two replace blocks swap values	Without temp one value is lost.
One pass sorts the list	It only moves the largest to the end.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
After one bubble pass, where is the largest value?

Homework
Sort 6 playing cards by hand using bubble passes. How many passes did you need?

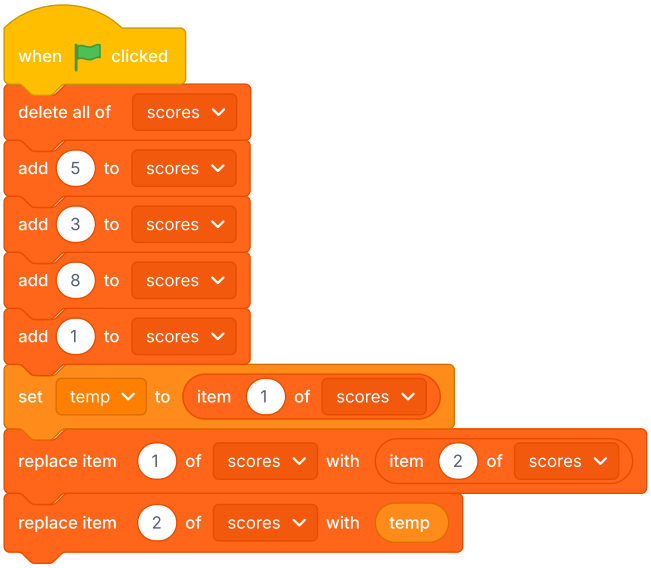
Answer key — Kitabash activities

1. Code Lab: swap two items (perform · code lab)

Goals checked: List is 3, 5, 8, 1 · Uses replace item · Uses temp

One solution:

Sorter



2. Debug the swap (perform · multiple choice)

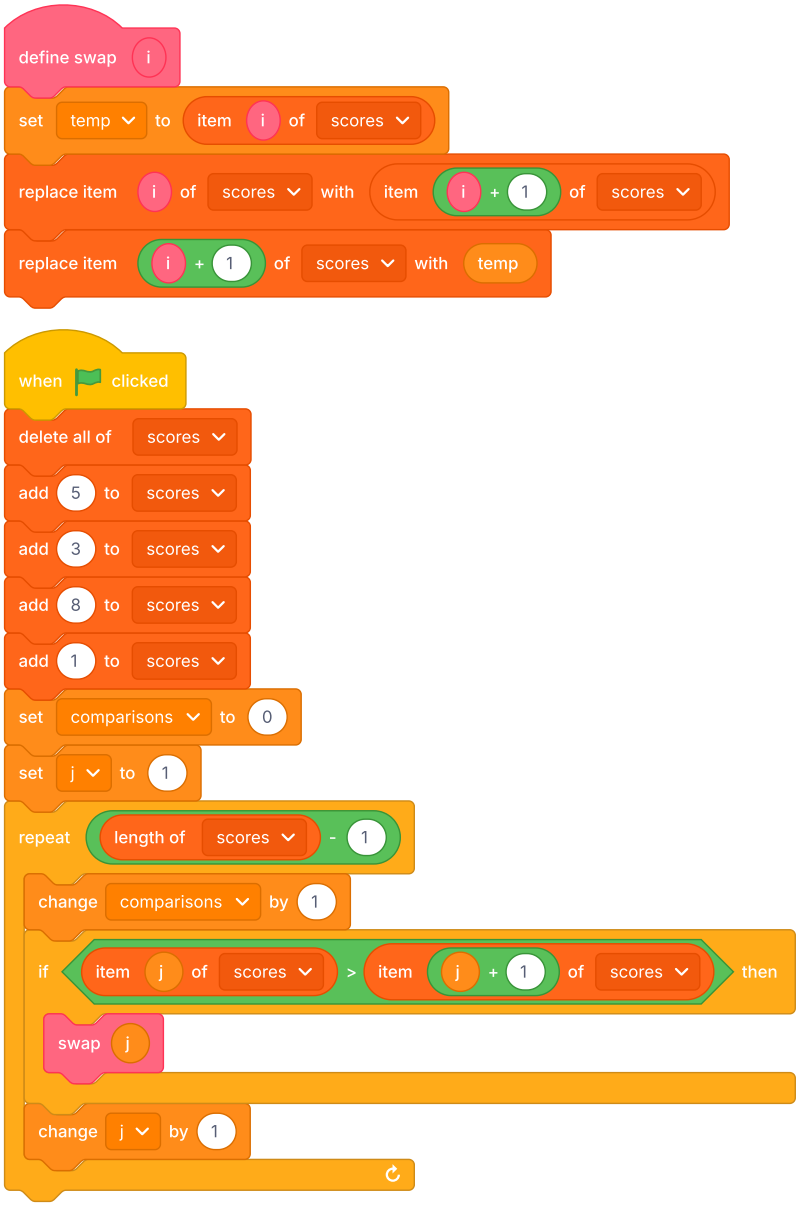
Answer: Item 1 was overwritten before it was copied — it needs a temp variable

3. Code Lab: one bubble pass (probe · code lab)

Goals checked: After one pass: 3, 5, 1, 8 · Uses swap · 3 comparisons

One solution:

Sorter



4. Explain your thinking (probe · free response)

Look for: temp, overwrite, swap. **Example:** If you copy item 2 into item 1 first, the old value of item 1 is overwritten and lost. A temp variable is like an empty cup: you pour item 1 into it, move item 2 across, then pour the temp back into position 2.

Answer key — worksheet

1. **Key words:** swap = Exchange two values. · temporary variable = A variable that holds a value briefly. · pass = One trip through the list comparing neighbours. · adjacent = Next to each other.
2. **Plan your code:** Goals: List is 3, 5, 8, 1; Uses replace item; Uses temp.

3. **Debug the swap:** A — After the first block, item 1 is already 3, so the 5 is lost.
4. **Plan your code:** Goals: After one pass: 3, 5, 1, 8; Uses swap; 3 comparisons.
5. **Explain your thinking:** If you copy item 2 into item 1 first, the old value of item 1 is overwritten and lost. A temp variable is like an empty cup: you pour item 1 into it, move item 2 across, then pour the temp back into position 2.

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